

SFO-X3855J-M:assignment5 csun\$ python3 assignment5.py

Part1: Imbalanced Data Set

Accuracy: 0.9417

Confusion Matrix:

```
[[40 0 0]
```

```
[ 0 23 0]
```

```
[ 0 7 50]]
```

Class balanced accuracy: 0.9308641975308642

Balanced accuracy: 0.9777777777777779

Scikit-learn balanced_accuracy_score: 0.9444444444444445

Part2: Oversampling

Random oversampling

Accuracy: 0.9733

Confusion Matrix:

```
[[50 0 0]
```

```
[ 0 46 0]
```

```
[ 0 4 50]]
```

SMOTE oversampling

Accuracy: 0.9667

Confusion Matrix:

```
[[50 0 0]
```

```
[ 0 45 0]
```

```
[ 0 5 50]]
```

ADASYN oversampling

Error occurred while executing ADASYN oversampling:

Not any neighbours belong to the majority class. This case will induce a NaN case with a division by zero. ADASYN is not suited for this specific dataset. Use SMOTE instead.

Part3: Undersampling

Random undersampling

Accuracy: 0.9333

Confusion Matrix:

```
[[30 0 0]
```

```
[ 0 28 4]
```

```
[ 0 2 26]]
```

Cluster undersampling

Accuracy: 0.9556

Confusion Matrix:

```
[[30 0 0]
```

```
[ 0 27 1]
```

```
[ 0 3 29]]
```

Tomek links undersampling

Accuracy: 0.958

Confusion Matrix:

[[40 0 0]

[0 25 0]

[0 5 49]]

```
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